

Worksheet — Real Numbers (Class 10 CBSE Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show ALL working. Total time: 50 minutes.

Q1. In Euclid's lemma $a = bq + r$, the remainder r satisfies: [1 mark]

- (A) $0 \leq r \leq b$
- (B) $0 < r < b$
- (C) $0 \leq r < b$
- (D) $r = b$

Q2. $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$ for any two positive integers a, b . [1 mark]

- (A) True
- (B) False

Q3. Prime factorisation of $60 = 2^2 \times \underline{\hspace{1cm}} \times 5$. [1 mark]

Q4. Does $7/12$ have a terminating decimal? [1 mark]

- (A) yes
- (B) no
- (C) depends
- (D) cannot tell

Q5. Find $\text{HCF}(96, 404)$ using prime factorisation. [2 marks]

- (A) 2
- (B) 4
- (C) 8
- (D) 16

Q6. Use Euclid's algorithm to find $\text{HCF}(420, 130)$. Show every step. [3 marks]

Q7. Find HCF and LCM of 24 and 36 by prime factorisation. Verify $\text{HCF} \times \text{LCM} = \text{product}$. [3 marks]

Q8. Find HCF and LCM of 12, 15 and 21. [3 marks]

Q9. Without performing the division, decide whether each has a TERMINATING or NON-TERMINATING decimal: (a) $7/8$, (b) $13/3125$, (c) $23/30$, (d) $64/455$. [3 marks]

Q10. Prove that $\sqrt{2}$ is irrational. [6 marks]

Q11. Three bells start ringing together at 7 a.m. They ring at intervals of 12, 15 and 18 minutes.
(i) When will they next ring together? (ii) How many times will they all ring together in the next 6 hours? [6 marks]

Answer Key

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Q1. In Euclid's lemma $a = bq + r$, the remainder r satisfies:

Answer: $0 \leq r < b$

Standard statement.

MCQ · EASY · 10-CBSE-MATH-RN-01

Q2. $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$ for any two positive integers a, b .

Answer: True

Standard identity.

TRUE_FALSE · EASY · 10-CBSE-MATH-RN-04

Q3. Prime factorisation of $60 = 2^2 \times \underline{\hspace{1cm}} \times 5$.

Answer: 3

$60 = 2^2 \cdot 3 \cdot 5$.

FILL_BLANK · EASY · 10-CBSE-MATH-RN-03

Q4. Does $7/12$ have a terminating decimal?

Answer: no

$12 = 2^2 \cdot 3$ — has a 3 — non-terminating.

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Q5. Find $\text{HCF}(96, 404)$ using prime factorisation.

Answer: 4

$96 = 2^5 \cdot 3$; $404 = 2^2 \cdot 101$. $\text{HCF} = 2^2 = 4$.

MCQ · MEDIUM · 10-CBSE-MATH-RN-04

Q6. Use Euclid's algorithm to find $\text{HCF}(420, 130)$. Show every step.

Answer: $420 = 130 \times 3 + 30$. $130 = 30 \times 4 + 10$. $30 = 10 \times 3 + 0$. Last non-zero remainder = 10. $\text{HCF}(420, 130) = 10$.

Three steps.

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Q7. Find HCF and LCM of 24 and 36 by prime factorisation. Verify $\text{HCF} \times \text{LCM} = \text{product}$.

Answer: $24 = 2^3 \cdot 3$; $36 = 2^2 \cdot 3^2$. $\text{HCF} = 2^2 \cdot 3 = 12$. $\text{LCM} = 2^3 \cdot 3^2 = 72$. Verify: $12 \times 72 = 864 = 24 \times 36$ ✓.

Use min/max powers; verify identity.

SHORT_ANSWER · MEDIUM · 10-CBSE-MATH-RN-04

Q8. Find HCF and LCM of 12, 15 and 21.

Answer: $12 = 2^2 \cdot 3$, $15 = 3 \cdot 5$, $21 = 3 \cdot 7$. HCF = 3 (only prime common to all). LCM = $2^2 \cdot 3 \cdot 5 \cdot 7 = 420$.

Three numbers.

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Q9. Without performing the division, decide whether each has a TERMINATING or NON-TERMINATING decimal: (a) $7/8$, (b) $13/3125$, (c) $23/30$, (d) $64/455$.

Answer: (a) $7/8$: $8 = 2^3$ — terminates. (b) $13/3125$: $3125 = 5^5$ — terminates. (c) $23/30$: $30 = 2 \cdot 3 \cdot 5$ — has 3 — NOT terminating. (d) $64/455$: $455 = 5 \cdot 7 \cdot 13$ — has 7 and 13 — NOT terminating.

Test denominators for $2^m \cdot 5^n$.

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Q10. Prove that $\sqrt{2}$ is irrational.

Answer: Assume $\sqrt{2}$ is rational; then $\sqrt{2} = p/q$ in lowest terms (HCF(p, q) = 1, $q \neq 0$). Squaring: $p^2 = 2q^2$. So p^2 is even, hence p is even (an odd integer's square is odd). Write $p = 2k$. Then $p^2 = 4k^2 = 2q^2$, so $q^2 = 2k^2$. So q^2 is even, hence q is even. But then both p and q have factor 2, contradicting HCF(p, q) = 1. So $\sqrt{2}$ is IRRATIONAL.

Standard contradiction proof.

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Q11. Three bells start ringing together at 7 a.m. They ring at intervals of 12, 15 and 18 minutes. (i) When will they next ring together? (ii) How many times will they all ring together in the next 6 hours?

Answer: (i) Need LCM(12, 15, 18). $12 = 2^2 \cdot 3$; $15 = 3 \cdot 5$; $18 = 2 \cdot 3^2$. LCM = $2^2 \cdot 3^2 \cdot 5 = 180$ min = 3 hours. Next together at $7 + 3 = 10$ a.m. (ii) After 7 a.m., they ring together at 10 a.m. and 1 p.m. (within the next 6 hours, until 1 p.m.). So 2 times.

LCM application.

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